Diseases of onion and garlic

1. Purple blotch - *Alternaria porri*
2. Downy mildew - *Peronospora destructor*
3. Onion smut - *Urocystis cepulae*
4. Onion smudge - *Colletotrichum circinans*
5. Neck rot/bulb rot - *Botrytis allii*, *B. squamosa*,*B. cinerea*
6. Onion blast - *Botrytis allii* , *B. byssoidea*, *B. squamosa*, *B. cinerea* and *B. cepae*
7. Stemphylium blight - *Stemphylium vesicarium*

Diseases of onion and garlic

Fungal Rots

1. Fusarium basal rot- *Fusarium oxysporum* f.sp. *cepae* and *solani*
2. White rot- *Sclerotium cepivorum*
3. Fungal Brown stain - *Botrytia cinerea*
4. Black mold/black rot - *Aspergillus niger*
5. Blue mold/blue Rot- *Penicilium italicum*.

Bacterial Rots

1. Soft rot- *Erwinia caratovora* pv *caratovora*
1. Slippery skin - *Pseudomonas allicola* pv. *allicola*
2. Sour skin - *Pseudomonas cepacia*
2. Bacterial Brown stain - *Pseudomonas aeruginosa*

Purple blotch - *Alternaria porri*

Purple blotch - *Alternaria porri*

- This disease is common in India, both in seed as well as in bulb crop and **mostly aggravated by *Botrytis* infection and thrips attack.**
- It is an important disease prevalent in all the onion growing areas.
- It is more common in kharif season

Symptoms:

- The characteristic symptoms of the disease appear as small, **water-soaked lesions turn brown with age**, and are usually **elliptical in shape on leaves or seed stalks**
- Similar lesions on seed stalks and **the seeds may not develop and shriveled.**

Purple blotch - *Alternaria porri*

Symptoms contd….

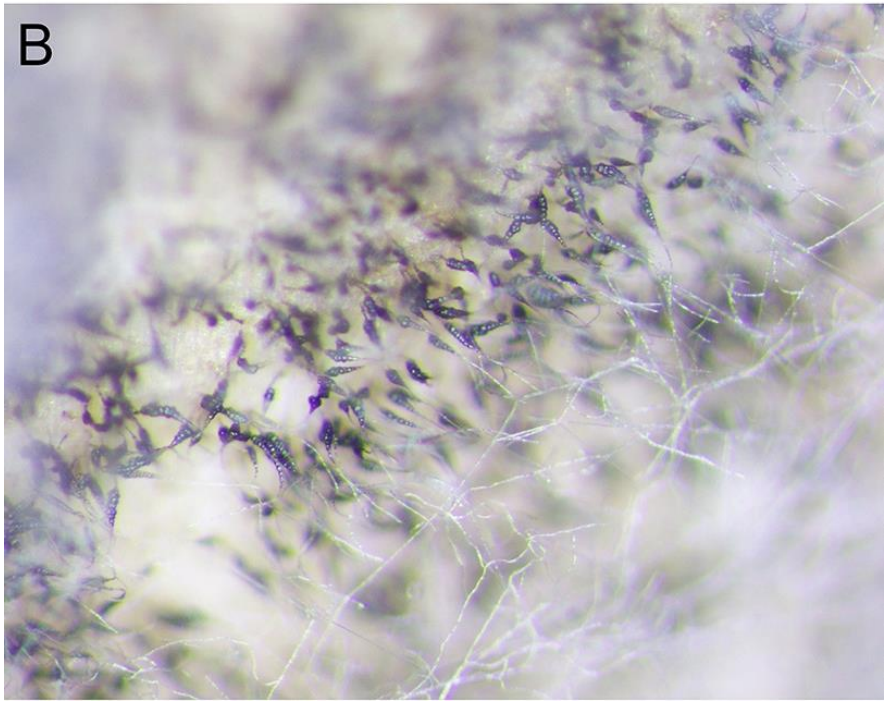
- Older purple blotch lesions appear to have **a target, zonate appearance and magenta pink-purple color**
- The lesion margin is often **a deep purple color surrounded by a halo.**
- In moist weather, the **surface of the spot is covered with the brown or almost black sporulation** of the fungus.
- Usually **the affected leaf or stem falls down and dies within 3 or 4 weeks** under favourable environmental conditions.
- **In garlic, the disease appears on leaves with the similar symptoms**.

Purple blotch - *Alternaria porri*

Pathogen:

- Conidiophores arise singly or in groups and are straight or flexuous, often **geniculate, septate, pale to mid brown in colour.**
- Conidia are **muriform and are usually solitary, straight or curved, obclavate** and taper to a beak that is commonly about the same length or slightly larger than the body of the conidium.
- Each conidium has 8 to 12 transverse and zero to several longitudinal or oblique septa, the **beak is flexuous, pale and tapering and each cell of the conidium is capable of germination by germ tube.**
- Existence of chlamydospores in the fungus is also reported.
- Sexual stage of the fungus is unknown.

Purple blotch - *Alternaria porri*



Purple blotch - *Alternaria porri*

Disease cycle and epidemiology:

Hot and humid climate with temperature ranging from 21-30°C and relative humidity (80-90%) favour the development of the disease.

- The fungus survives from one season to other in infected plant debris as dormant mycelium.
- The fungus in diseased onion leaf and seed stalk debris can also **survive for 12 months buried at 5 and 7.5 cm depth**.
- Wherever the chlamydospores of the fungus are formed they can also serve as source of perennation.
- Persistent dew for extended periods or high relative humidity (>85%) are essential for infection and sporulation.
- The fungus can grow over a wide range of temperature (6-34° C) with optimum at 25°C.

Purple blotch - *Alternaria porri*

Management:

- Collect and destroy the infected plant debris. Follow crop rotation with unrelated crops.
- Cvs./ lines **viz. Pusa Red, IIHR-56-1, IC 48059, IC 48179, IC 39887, IC 48025, ALR and Ro-1** have been reported as resistant to this disease.
- Treat the seed with captan (0.3%) or thiram (0.3%).
- With the initiation of the disease, spray the crop with mancozeb (0.25%) or copper oxychloride (0.3%) or hexaconazole (0.05%) and repeat at 10-14 days interval.

Stemphylium blight - *Stemphylium vesicarium* (PS : *Pleospora allii*)

Stemphylium leaf blight

Stemphylium blight - *Stemphylium vesicarium* (PS : *Pleospora allii*)

Stemphylium blight - *Stemphylium vesicarium* (PS : *Pleospora allii*)

Symptoms

- The **Stemphylium blight is a serious problem in Northern parts of the country especially in the seed crop.**
- This disease is **very common on onion leaves and flower stalks.**
- Infection occurs on leaves of transplanted seedlings at **3- 4 leaf stage during late March and early April.**
- Lesions are initially small, **tan to brown**, and may be water-soaked, becoming **spindle shaped to ovate-elongate or diffused spots**

Stemphylium blight - *Stemphylium vesicarium* (PS : *Pleospora allii*)

Symptoms contd…

- As lesions age, the centers progress from **light brown to tan in color to dark olive brown and black** due to the production of conidia.

- Lesions may be surrounded by **a light pink halo**

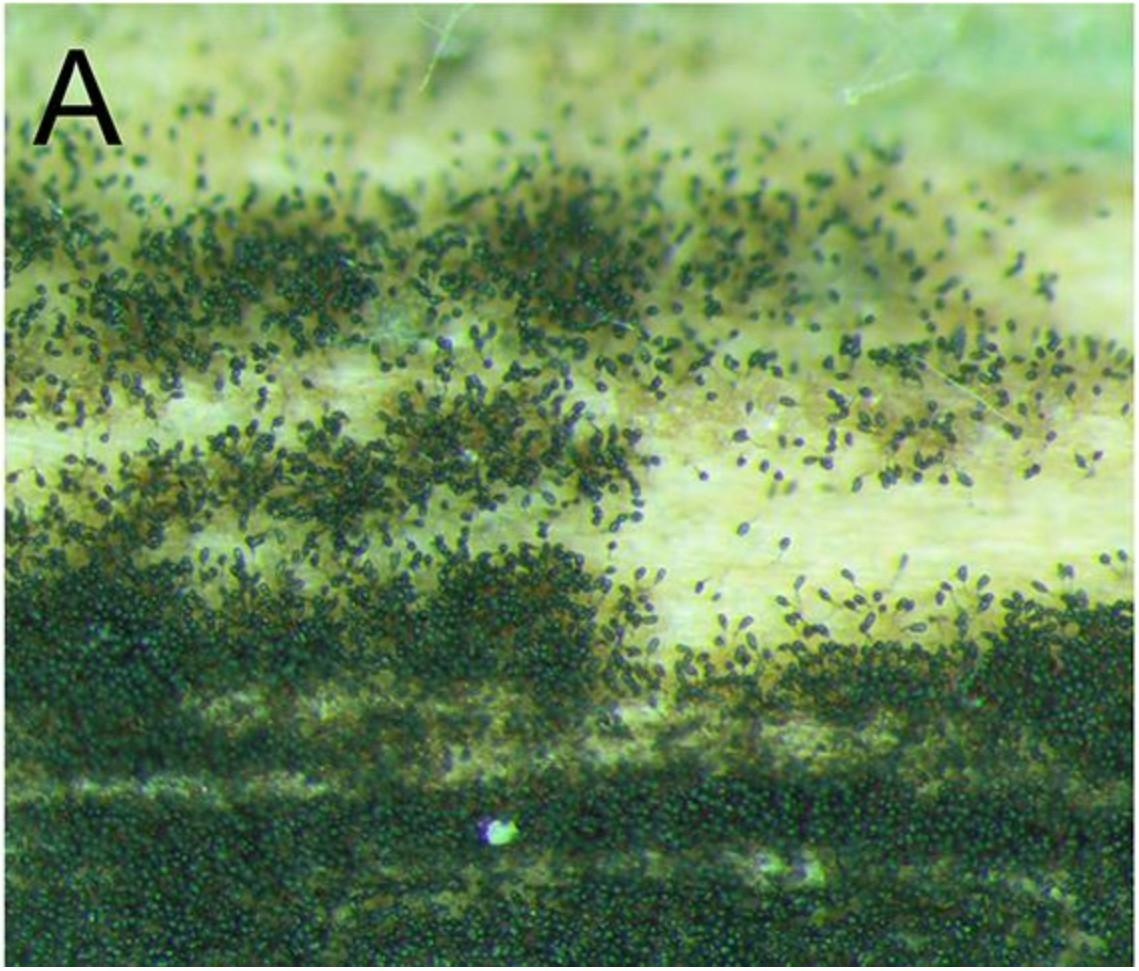
- As the disease spreads, the lesions rapidly progress along the leaf initially in an asymmetric pattern causing longer lesions down one side of the leaves but eventually resulting in complete defoliation.

Stemphylium blight - *Stemphylium vesicarium* (PS : *Pleospora allii*)

Symptoms contd…

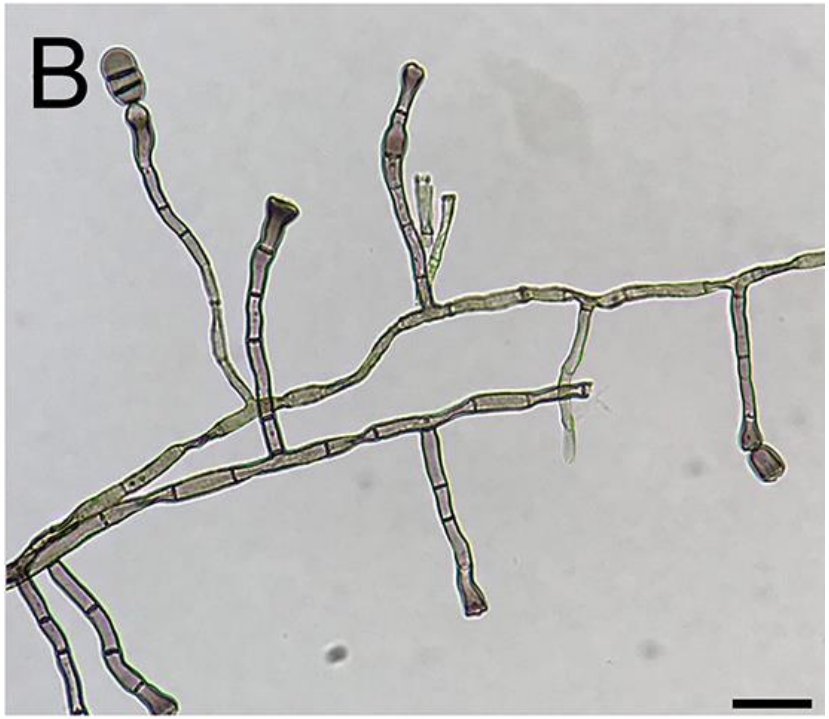
- Signs of the pathogen (production of conidia) are commonly observed on infected tissues.
- *S. vesicarium* colonization of necrotic tips and other tissue affected by dieback is commonly referred to as **'dirty tips'.**
- Similar symptoms appear on the inflorescence stalks.
- In garlic crop, **the symptoms of disease appear on the inner side of the leaf only.**

Stemphylium blight - *Stemphylium vesicarium* (*PS : Pleospora allii*)



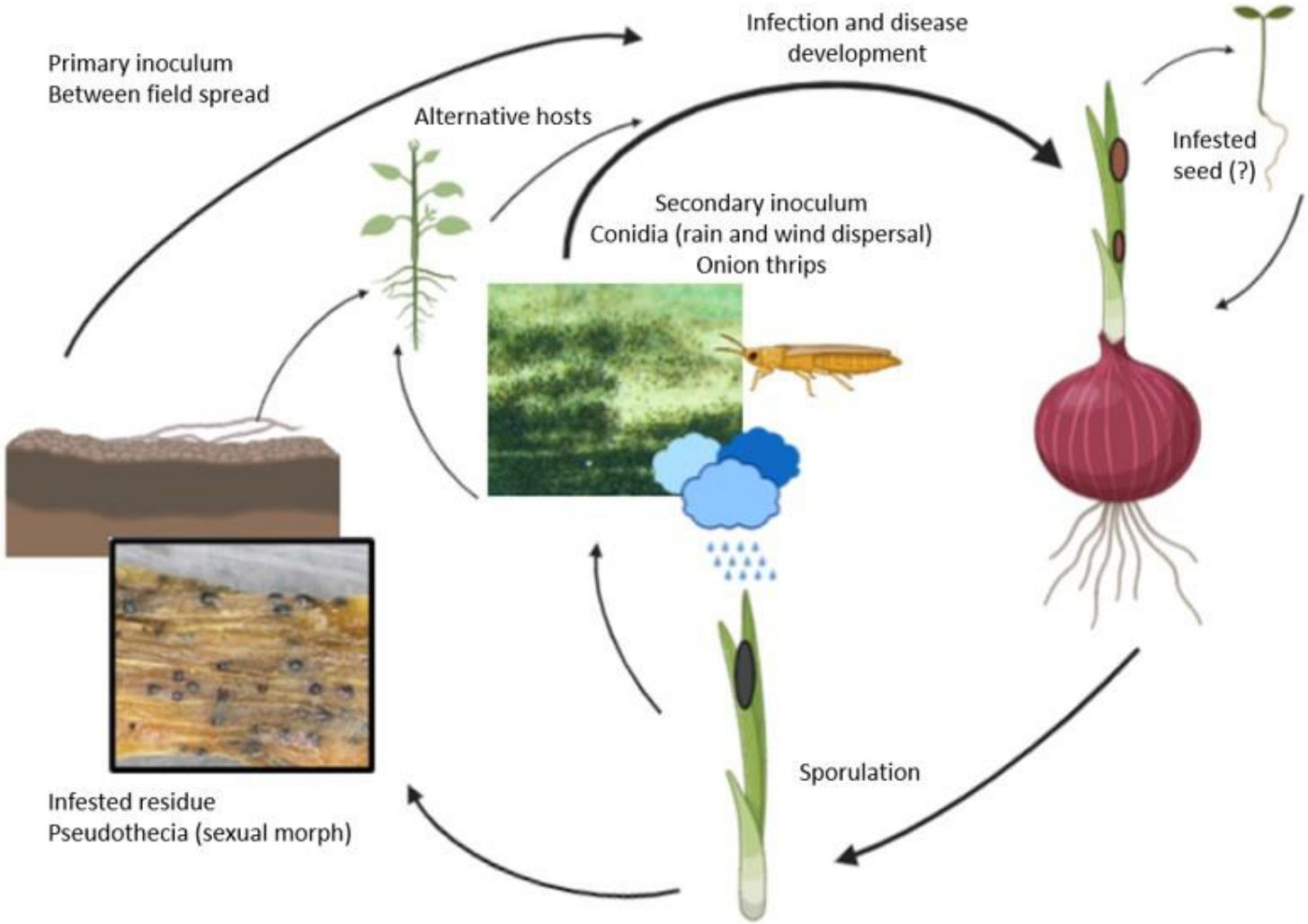
A . *S. vesicarium* - profuse sporulation on lesions of onion leaves.

Stemphylium blight - *Stemphylium vesicarium* (*PS : Pleospora allii*)



(B) Conidiophore (C) Broadly oval to block-shaped conidia

Stemphylium blight - *Stemphylium vesicarium* (*PS : Pleospora allii*)



Stemphylium blight - *Stemphylium vesicarium* (PS : *Pleospora allii*)

Pathogen:

- Conidiophores are **straight to variously curved, 1-4 septa,** simple or occasionally one branched, cylindrical but enlarging apically to the site of the conidium production, light yellow brown to medium golden brown in the swollen apex.
- Conidia are **oblong or broadly oval** and sometimes in equilateral one to six transverse septa and one to three longitudinal septa and constricted at the major transverse septa.

Stemphylium blight - *Stemphylium vesicarium* (*PS : Pleospora allii*)

Pathogen:

- Conidia are light to medium golden brown to olive brown in colour and each has **a conspicuous basal scar like zone.**
- Perithecia mature within 3 to 6 months.
- Asci are cylindrical to clavate in shape.
- Ascospores are ellipsoidal and the upper half is narrowly tapered.

Stemphylium blight - *Stemphylium vesicarium* (*PS : Pleospora allii*)

Disease cycle and epidemiology:

- The fungus perpetuates in the infected plant debris.
- Moderate temperature and humidity coupled with precipitation favour the disease initiation and development.

Stemphylium blight - *Stemphylium vesicarium* (*PS : Pleospora allii*)

Management:

- Cultural practices like collection and destruction of plant debris, crop rotation and summer ploughings are recommended for the control of this disease.

- Sprays of mancozeb (0.25%) or chlorothalonil or carbendazim or propineb or difenconazole along with sticker Triton or Sandovit have been found to be very effective for the control of this disease.

- Initiate sprays in the first week of February and repeat at 10-14 days interval.

Downy mildew - *Peronospora destructor*

Downy mildew - *Peronospora destructor*

Symptom development on an affected scape.

Sporulation on an infected leaf.

Downy mildew - *Peronospora destructor*

Peronospora destructor sporulation on leaves.

Extensive foliar damage in the field.

Brownish-purple sporulatio on healthy green leaves.

Peronospora destructor infection may be followed by invading secondary organisms, often leading to plant collapse.

Downy mildew - *Peronospora destructor*

- This disease was first time reported in Britain (1841), then in USA (1844), whereas in India it was first recorded during 1974-75 from Kashmir valley and appeared destructive in the year 1975-76.

- Yield losses upto 60-70 per cent have been reported.

Symptoms:

- Two types *i.e.*, **local and systemic infection occur.**
- **Systemic infection occurs** when the plants are raised from **diseased bulbs or infected seedlings** are used for planting.
- Plants raised from such bulbs **remain stunted**, become **distorted and light green in colou**r
- In humid weather conditions, **felty whitish to grayish fungal growth with sporulation.**

Downy mildew - *Peronospora destructor*

Symptoms contd..

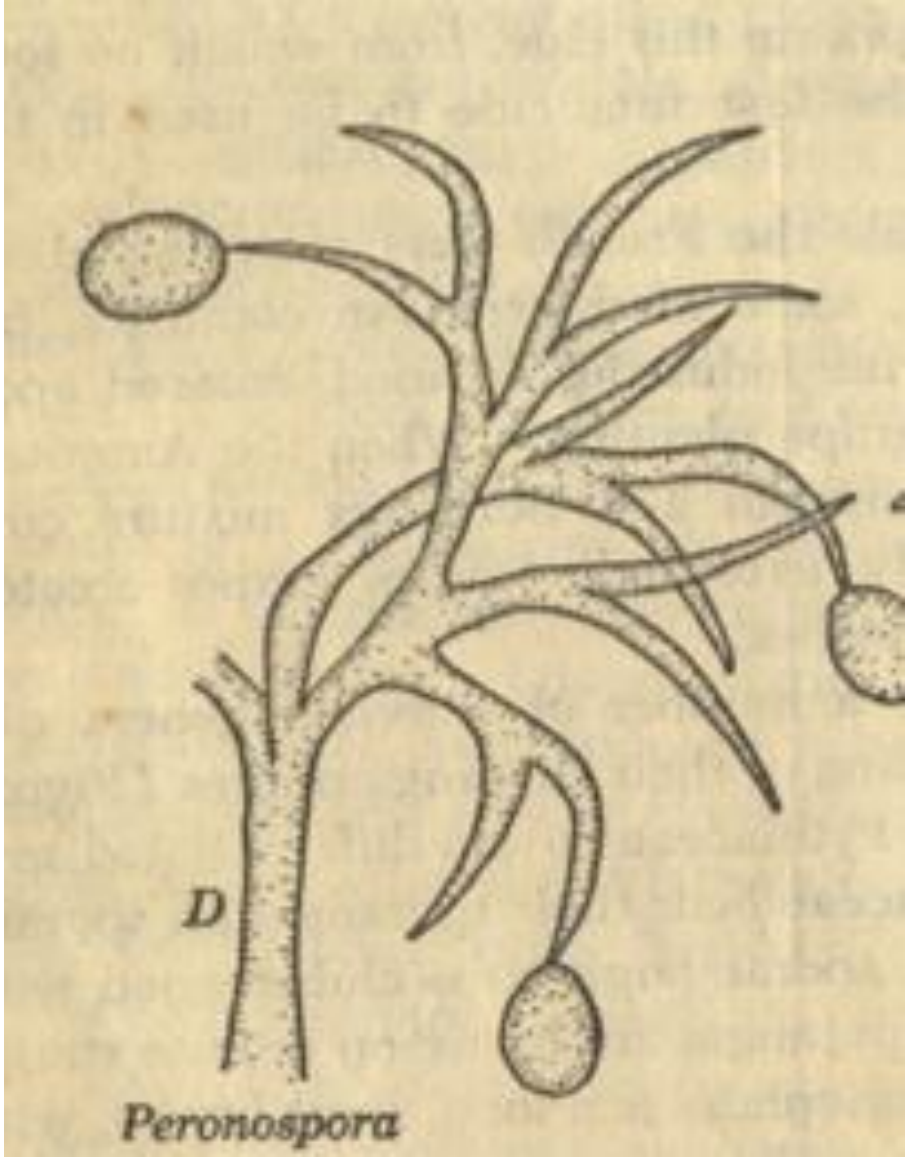
- Local infection by air borne zoopsores which produce **oval to cylindrical spots.**
- In humid weather, the fungus develops as **white to purplish downy growth** on these spots.
- Affected **leaves become pale green, fold over, later collapse** and in such cases undersize bulbs are produced.
- On seed stalks, **circular to elongate lesions are produced and infected stalks** break over with the weight of the seed umbel, thereby causing the seed to shrivel.
- The disease is worst in damp conditions and late planting of the crop, application of **higher doses of fertilizers and numerous irrigations increase disease severity**.

Downy mildew - *Peronospora destructor*

Pathogen: *Peronospora destructor*

- The mycelium of the fungus is non-septate, intercellular with filamentous haustoria.
- The sporangiophores are aseptate, hyaline swollen at the base and 3 to 4 times dichotomously branched.
- The sterigmata were acute to subacute and bear pyriform to fusiform sporangia.
- Sporangia behave as conidia
- Oospores develop late in the season and germinate by germ tube.

Peronospora spp



Downy mildew - *Peronospora destructor*

Disease cycle and epidemiology:

- The main source of primary infection is mycelium in infected onion bulbs and such bulbs when used for seed production give rise to diseased plant.
- The fungus may also perpetuate as oospore.
- The disease spread by wind borne, and short lived spores.
- **The onion seedlings from disease prone areas also play significant role in introduction of disease in new localities.**
- The sporulation and infection by *P. destructor* usually takes place overnight while dissemination of the newly formed conidia occurs during morning hours.
- The temperature of 13° C and > 95 per cent relative humidity are optimum for the development of the disease.

Downy mildew - *Peronospora destructor*

Disease cycle and epidemiology contd:

- Presence of dew or raindrops on the leaf surface further enhances the chances of infection.
- Closely spaced and densely planted crops develop more disease compared with widely spaced ones while increase in nitrogen fertilization increases disease severity proportionately and potassium decreases .
- **A preliminary model, ONIMIL was developed to forecast primary infection,** which is able to determine for each day the probability of *P. destructor* establishing an infection on onion and its infectivity level.

Downy mildew - *Peronospora destructor*

Disease cycle and epidemiology contd:

MILIONCAST, a model was developed based on the data from the controlled environment studies to predict the rate of sporulation in relation to temperature and relative humidity.

- Sporangia were produced most rapidly at 8- 12°C after 5 h of high humidity during dark periods.
- The greatest number of sporangia was produced at 100% relative humidity, while no production at 93% .

Downy mildew - *Peronospora destructor*

Management:

- Collection and burning of the infected plant debris.
- Use healthy seed/ planting material for propagation.
- **Onion lines like IC 48045, IC-32149, IC-33617, IC-49371 and DOP-11 have been reported to be resistant.**
- Spray the crop with metalaxyl + mancozeb (0.25%) immediately with the initiation of the disease followed by sprays of mancozeb (0.25%) or copper oxychloride (0.3%) and repeat at 10-14 days interval.

Onion smut - *Urocystis cepulae*

Figure 1. Onion smut on young seedlings (courtesy

Onion smut - *Urocystis cepulae*

Early symptoms of smut infection manifest as black streaks on leaves.

Onion smut - *Urocystis cepulae*

Onion smut - *Urocystis cepulae*

Symptoms:

- Since the fungus remains in soil, disease appears on cotyledon of the young plant soon after it emerges.
- On the **cotyledons as dark, thickened areas near the base of seedlings** involving one to several millimeter of the surface.
- The **seedlings often die before emergence**.
- Large lesions can cause leaves to curve downward
- Mature lesions exposed, black, powdery spore masses (teliospores).

Onion smut - *Urocystis cepulae*

Symptoms contd…

- Infection progresses inward from leaf to leaf and infected plants become **stunted and may die within 3 to 4 weeks after emergence.**
- **If the plant survives, the disease becomes systemic and the plant remains in vegetative phase for the whole growing season.**
- **Bulbs are also covered with blackish lesions**.

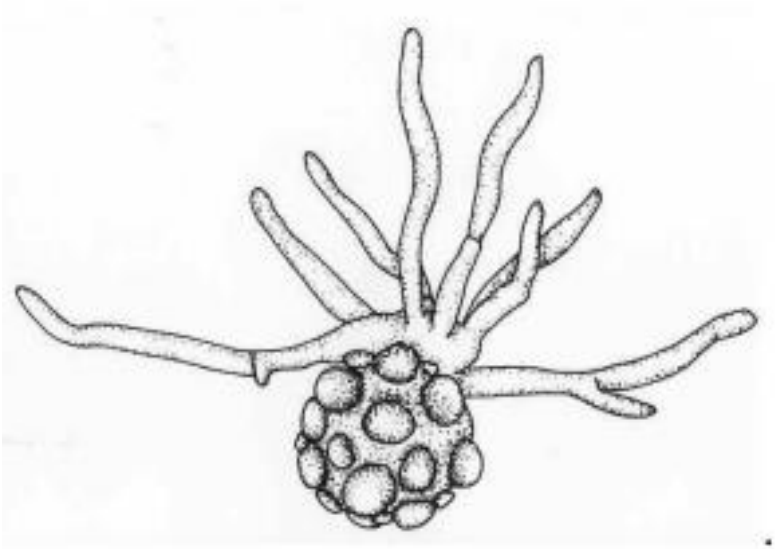
Onion smut - *Urocystis cepulae*

Pathogen:

- The disease is caused by *Urocystis cepulae*
- The sori of fungus looks dark coloured spore masses.
- Spore balls **consists of 1-4 fertile central cells, spherical or ellipsoid, reddish brown, smooth, thick walled, surrounded by a layer of small, slightly coloured, sterile cells** and fertile cells germinate by means of short promycelium while still held in ball.

Onion smut - *Urocystis cepulae*

- The sori of fungus looks dark coloured spore masses.
- Smut teliospores are single celled, spherical or ellipsoid, reddish brown, smooth thick walled



Onion smut - *Urocystis cepulae*

Disease cycle and epidemiology:

- The pathogen is soil borne and perpetuates in the form of chlamydospores.
- If the host is available, the spores germinate immediately otherwise they remain dormant indefinitely.
- Optimum temperature for spore germination and growth is 13 to 22 ^{o} C.

Onion smut - *Urocystis cepulae*

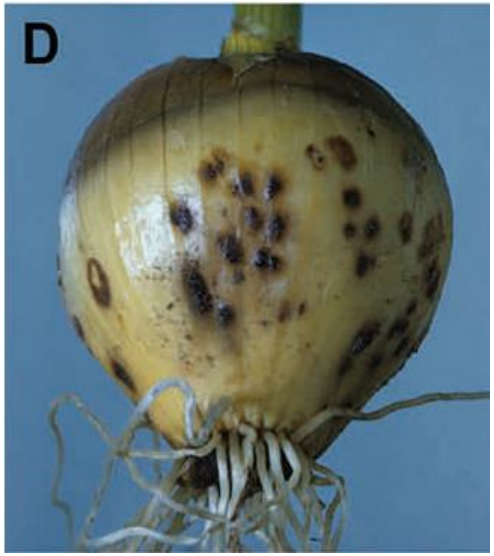
Management:

- Use healthy seed and treat with thiram or captan (0.3%).
- Before sowing, the nursery bed should be treated with Formaldehyde (8 ml of 40% formaldehyde/lit water) **@** 1 lit solution/30 feet furrow.
- Avoid the raising of nursery at same location every year.
- Cultivar- **Hardy is resistant**

Onion smudge - *Colletotrichum circinans*

Colletotrichum blight

Curling and abnormal elongation of leaves



Concentric rings of fungal fruiting bodies form on the surface of bulb scales.

Onion smudge - *Colletotrichum circinans*

- Walker (1929) gave biochemical resistance *i.e.* red onion cv. posses **catechol** and protocatechuic acid that provide resistance against onion smudge pathogen.

- This disease is more prevalent in **white coloured onion var. grown in temperate region of the world**. **It is one of the important diseases in storage.**

Symptoms:

- In seed bed it causes damping off of seedlings.

- Disease is characterized by the appearance of dark green to black smudge (minute stromata of the fungus) on bulb, neck or green leaves.

- In humid climate pinkish masses of spore can be seen.

Onion smudge - *Colletotrichum circinans*

Pathogen - *Colletotrichum circinans*

- Acervuli are formed on stromata just beneath the cuticle by formation of pallisade layer of short conidiophores along with dark satae.
- Conidia are fusiform, falcate and hyaline.

Onion smudge - *Colletotrichum circinans*

Disease cycle and epidemiology:

- Pathogen survives in stromata and as a saprophytic mycelium in the debris.
- Stromata give rise to acervuli and conidia.
- Disease development takes place at 10- 32ºC but optimum being 26ºC.
- Moist condition favours the production of conidia.

Onion smudge - *Colletotrichum circinans*

Management:

- Grow coloured variety to minimize disease incidence.
- Dry the bulbs before storage for curing.
- Foliar application of mancozeb (0.25%) in combination with carbendazim (0.05%) will reduce disease development.

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*

Figure 3: Early foliage symptoms of blast.

Several Examples of
Botrytis Leaf Blight in Onions

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*

Symptoms:

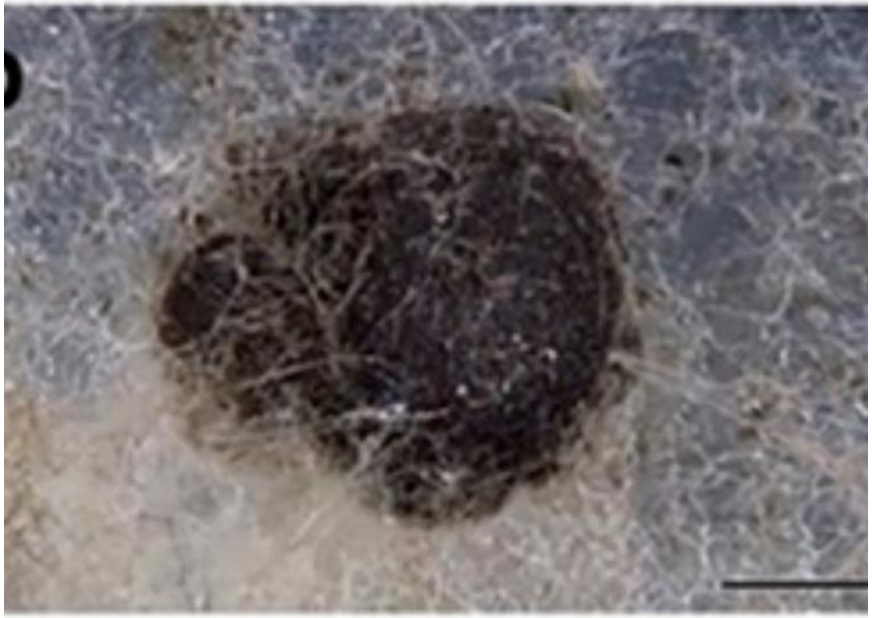
- The symptoms of the disease appear as **white specks with necrotic centres** surrounded by **a light green halo and lesions** may be either isolated and few in number or numerous
- They may expand slightly with age and take on an **elliptical shape and the halo** may disappear.
- Numerous lesions remain restricted in size on a single leaf result in dieback of the entire onion top, giving severely affected fields `**blasted' appearance**.
- Such leaves are killed prematurely resulting in reduction of bulb yield.

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*

Pathogen:

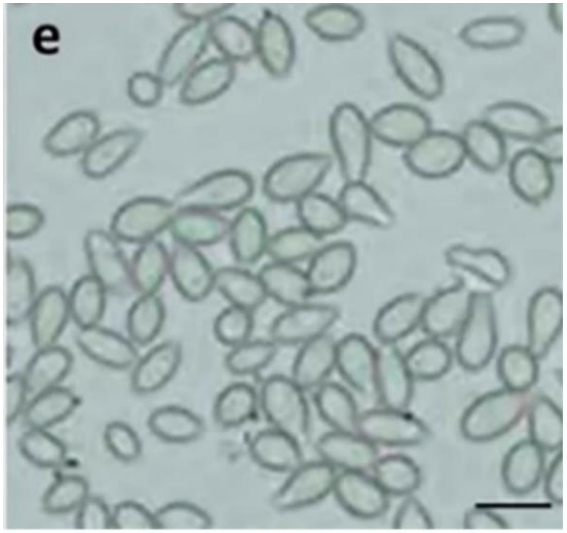
- The disease is caused by several species of *Botrytis allii*, *B. byssoidea*, *B. squamosa*, *B. cinerea* and *B. cepae*
- Botrytis is characterized by brown, erect , simple variable in length become hyaline and irregularly branched at the apex
- Ultimate branches bear densely aggregated conidia, each conidium being borne on minute sterigmata
- Conidia are hyaline or tinted, single celled and globose to ovoid.
- Sclerotia may form on leaf debris or on the necks of onion bulb and germinate by forming stipes on which conidiophore and conidia are produced.

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*



b. Sclerotia
c. hyphae,

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*



d.Conidiophore
e. Conidia.

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*

Disease cycle and epidemiology:

- The fungus perpetuates as sclerotia or mycelia in crop debris.
- Sclerotia upon germination produce conidia at 3 to 27° C (optimum at 9° C) and serve as source of primary inoculum.

Onion blast - *Botrytis allii , B. byssoidea, B. squamosa, B. cinerea, B. cepae*

Management:

- Collect and destroy the infected plant debris.
- Follow crop rotation avoiding Allium spp. in rotation.
- With the initiation of the disease, spray the crop with captan (0.25%) or mancozeb (0.25%) and repeat at 10-14 days interval.

Neck rot/bulb rot - *Botrytis allii, B. squamosa, B. cinerea*

Bulb longitudinal-section showing early symptoms of neck rot.

Bulb longitudinal-section showing advanced symptoms of neck rot.

Neck rot/bulb rot - *Botrytis allii, B. squamosa, B. cinerea*

Soil-line neck rot development on a bulb.

Botrytis allii mycelia and sclerotia on a bulb.

Neck rot/bulb rot - *Botrytis allii, B. squamosa, B. cinerea*

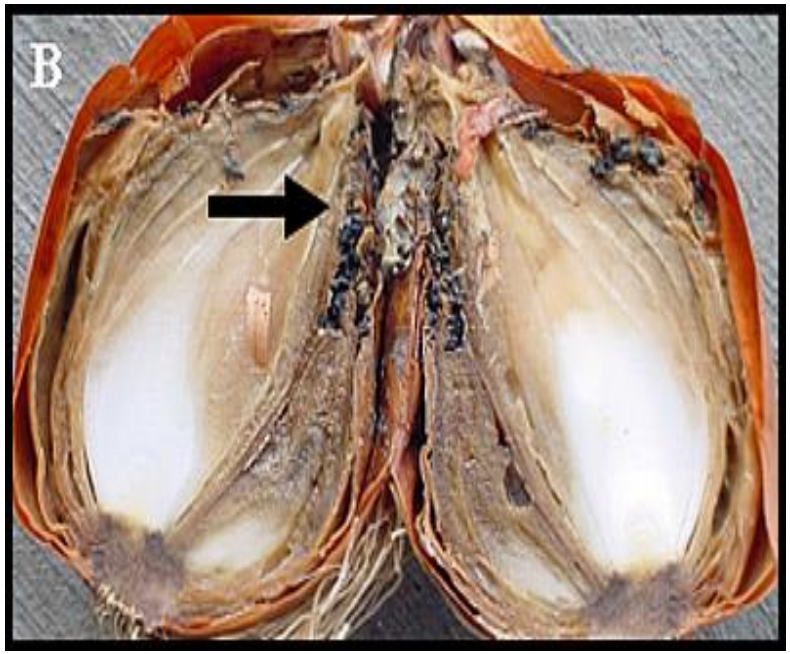


Figure 4: Botrytis infected cull onion. Note gray mold and black sclerotia.

Neck rot/bulb rot - *Botrytis alii, B. squamosa, B. cinerea*

Symptoms

- Symptoms starts at neck region at the time of harvest.
- The neck region becomes sunken and dried out, due to softening of the affected tissue
- IF the infected bulb is cut, dense layer of brown, gray or black mold appears at the neck.
- The infection progresses most rapidly down the scale which have been originally infected.

Neck rot/bulb rot - *Botrytis alii, B. squamosa, B. cinerea*

Symptoms contd…

- In advanced stage the bulb will shrivel and become covered with grey fungal growth
- Dark sclerotia appears on the older decayed tissue.
- Finally the entire bulb covered with brown mycelium.

Neck rot/bulb rot - *Botrytis alii, B. squamosa, B. cinerea*

Management

- Avoid excessive and late applications of nitrogen.
- Provide good ventilation for curing of onions before storage.

Fusarium basal rot- *Fusarium oxysporum* f.sp. *cepae* and *F. solani*

Symptoms:

- The main symptoms of the disease are wilting and rapid dying back of leaves from the tips **as the plant approach maturity.**

- In early stage of infection, the roots become pink in colour and rotting take place later

- In advanced stage, the bulb starts decaying from lower ends and ultimately whole plant die.

Fusarium basal rot- *Fusarium oxysporum* f.sp. *cepae* and *F. solani*

Symptoms contd….

- Infected garlic plant shows reddish or reddish purple discolouration on stems and bulbs early in the season
- Discolouration on bulb sheath at harvest
- **The bulbs become soft and when cut a watering decay is noticed**
- The **damping-off symptoms of seedlings** are also noticed.

Fusarium basal rot- *Fusarium oxysporum* f.sp. *cepae* and *F. solani*

Pathogen:

- The disease is caused by *Fusarium oxysporum* f.sp. *cepae* and *F. solani*
- Both species of *Fusarium* produces microconidia, macroconidia and chlamydospores.
- Macroconidia are uniformly curved and have 3-4 septa.

Fusarium basal rot- *Fusarium oxysporum* f.sp. *cepae* and *F. solani*

Disease cycle and epidemiology:

- Both pathogens perpetuate in soil through chlamydospores.
- The pathogen (s) may have been disseminated widely by infected onion sets and garlic cloves.
- The disease generally appears when soil temperature is 25-28° C along with high soil moisture.

Fusarium basal rot- *Fusarium oxysporum* f.sp. *cepae* and *F. solani*

Management:

- Follow long crop rotation because both pathogens are persistent soil inhabitants.
- Cultivar like IIHR Yellow, breeding line SI.29 and Hybrid 1 have some level of resistance to this disease.
- Dipping of seedlings before transplanting in the suspension of carbendazim (0.1%) is recommended.
- **Pre-harvest sprays of carbendazim** (0.1%) also reduce the post harvest decay in storage.

White rot - *Sclerotium cepivorum*

Symptoms:

- The fungus invades **roots and the basal part of the bulb** scales.
- The first symptoms of the disease appear as **yellowing and wilting of the leaves** followed by a total collapse of the tops.
- If the diseased plants are pulled gently, they will come up readily because of the rotting of centre roots and **affected parts are covered with a thick white mycelial** mat bearing numerous small sclerotia

White rot - *Sclerotium cepivorum*

Pathogen:

- The disease is incited by *Sclerotium cepivorum* .
- The fungus produces sclerotia which are uniformly round measuring 0.35 to 0.50 mm in size.
- Sclerotia germinate only once.

White rot - *Sclerotium cepivorum*

Disease cycle and epidemiology:

- The fungus is mainly disseminated by bulbs which have come from infected soil.
- The fungus persists in the soil and remains viable indefinitely.
- Disease develops rapidly in the soil having 40 per cent water holding capacity.
- Sclerotial germination occurs at temperatures ranging from 9 to 24° C with optimum being 14 to 18° C with low moisture.

White rot - *Sclerotium cepivorum*

Management:

- Collect and destroy the infected plant debris.
- Removal of infected plants during season reduces the sclerotial population and also avoids incorporation of the same in the soil.
- Soil solarization during summer months also reduces the incidence of this disease.
- Incorporation of *Trichoderma harzianum* in soil after solarization also controls *S. cepivorum* in soil effectively(40 g/m2).
- Several fungicides have been advocated but Iprodione was found most effective in containing this disease.

Black mold (*Aspergillus niger*)

Black mold/black rot - *Aspergillus niger*

Symptoms

- The disease is common in onions **stored in hot climates** where the temperature ranges between **30- 45°C.**
- **Infection usually is through neck tissues as foliage dies down at maturity.**
- Infected tissues become water soaked
- Initial **white mould formed between scales** later this growth produce black spores on stalks
- Development of black colour dots on the inner scales that arrange longitudinally.

Blue mold - *Penicillium* spp.,

Blue mold/blue rot - *Pencilium italicum*

Symptoms

- Initially pale yellow, watery spots appear on the onions
- Later turned into **bluish green fungal growth.**
- If the onion is cut open it can be seen scales appear **watery and brownish grey**
- **After a while the bulbs become soft and wet rot can develop**
- Development of blue colour spores on the inner scales.
- **Infected bulb gives musty smell.**

Blue mold - *Penicillium* spp.,

Management and control

- Avoid damage to the bulbs during harvesting, storage and transport
- **Store the bulbs at <15^{0}C and <75%RH**

Fungal Brown stain - *Botrytia cinerea*

Brown stain on the neck and outer bulb scales.

Fungal Brown stain - *Botrytia cinerea*

Symptoms

- Spores of the fungus germinate **on onion leaves** and produce enzymes that result in **superficial flecking**.
- When the fungus **grows into the bulb scales, it causes a brown stain on the neck and outer scales.**

Fungal Brown stain - *Botrytia cinerea*

Management

- Harvest at the correct time at least 50% of the foliage has died
- Avoid damaging or injuring at the harvesting time.
- Sort out and **remove damaged or thick necked bulbs** from the lot before storing.
- Periodical removal of rotted bulbs during storage.
- **Dry the onions as quickly as possible after harvesting**
- Storage of bulbs **at <15^{0}C and <75%RH.**

Fungal Brown stain - *Botrytia cinerea*

Management

- Bulb treatment with Thiram, Captan, Vinclozolin 4 g/kg or Carbendazim 2 g/kg.
- Spraying vinclozolin 0.1% or carbendazim 0.1%.

Bacterial soft rot - *Erwinia caratovora* pv *caratovora*

Figure 18. Deterioration of the core bulb scales caused by bacterial soft rot.

Bacterial soft rot - *Erwinia caratovora* pv *caratovora*

Symptoms

- *Erwinia* can infect the plants **in field**
- Bacterial soft rot is mainly a problem **on mature bulbs.**
- Formation of water soaked lesions at the base of the plant, drying of infected leaves.
- Affected scales first appear **water soaked and pale yellow to light brown.**
- Disease spread to the bulbs.

Bacterial soft rot - *Erwinia caratovora* pv *caratovora*

Symptoms

- As the soft rot progresses, invaded fleshy **scales become soft and sticky, watery** with the interior of the bulb breaking-down.
- **Infected bulbs shows watery, sticky appearance**
- A watery, **foul-smelling thick liquid can be squeezed** from the neck of diseased bulbs.

Slippery skin - *Pseudomonas alliicola* pv. *alliicola*

Figure 14. Collapsed leaves caused by bacterial

Figure 15. Dark green lesions caused by bacterial

Slippery skin - *Pseudomonas alliicola* pv. *alliicola*

Symptoms

- Field symptoms often appear as **one or two wilted leaves in the center of the leaf cluster**.
- These leaves eventually turn pale **yellow and dieback from the tip** while older and younger leaves maintain a healthy green appearance.
- **Softening of the** neck tissue.
- In a longitudinal section, one or more inner **scales become watery.**
- Eventually, all the internal **tissue will rot.**

Slippery skin - *Pseudomonas alliicola* pv. *alliicola*

Symptoms contd…

- Affected scales later turn brown then bulbs dry out and shrivel.
- **Squeezing the base of infected plants causes the rotted inner portion of the bulbs to slide out** through the neck, hence the name slippery skin.
- Finally, the internal scales dry and the bulb shrivel (Garlic)
- Infected bulbs turn yellow colour.

Sour skin - *Pseudomonas cepacia*

Sour skin - *Pseudomonas cepacia*

Symptoms

- In contrast to soft rot and slippery skin, **infected scales are not water-soaked but are slimy and yellow**
- **Young foliage is very susceptible** to this disease.
- Leaves are turn light brown
- **Outer scale becomes light yellow, later producing a yellowish brown slime**
- Infected bulbs **give off a typically sour smell/ vinegar-like odour**

Management of bacterial diseases

- Onion tops should be allowed to mature well before harvesting.
- Care should be taken to avoid damage during harvesting and packing.
- Proper curing and drying at harvest time
- Discard all infected bulbs before storing
- Storage places should be well ventilated to avoid accumulation of moisture on the surfaces of bulbs.
- Periodical elimination of rotted bulbs
- Onions should be stored at 0°C and a relative humidity of 65-70%.